

Supplementary material

Short-interval test–retest reliability of artificial intelligence–derived ECG age and sex

All values are computed from aggregate outputs of the analysis code. Abbreviations: S_w , within-patient standard deviation; RC, repeatability coefficient ($2.77 \cdot S_w$); LoA, 95% limits of agreement; ICC, two-way random-effects absolute-agreement single-measure intraclass correlation; MAE, mean absolute error; RBBB/LBBB, right/left bundle branch block. Cohorts: I0001, HEEDB I0001 test patients (Massachusetts General Hospital); I0006, HEEDB I0006 (Emory University Hospital); MIMIC, MIMIC-IV-ECG; MIMIC ED, troponin-negative emergency subcohort.

Supplementary Methods

S1. Pair construction and signal quality

For each patient, ECGs were ordered by acquisition time and record identifier, and each record was paired with the next record of the same patient. Pairs with identical acquisition times were discarded. Each pair was assigned to one of eight interval strata. Within a stratum, one pair per patient was kept by the smallest hash of the concatenated record identifiers, and at most 25 000 patients per stratum and cohort were kept by the smallest hash of the patient identifier. Both hashes are deterministic, so the sample is reproducible. In HEEDB I0001, only patients in the database's test split were used; models were trained only on training-split patients.

Signals were read in physical units (mV) with the WFDB library. A record was excluded if: - the 12 standard leads, a 10-second duration (± 0.05 s), or millivolt units were missing; - any sample was non-finite; - any lead had an SD below 0.005 mV; - any absolute amplitude exceeded 20 mV.

The lead-wise median was subtracted. For the in-house models, signals were resampled to 250 Hz (2500 samples). For the published models, they were resampled to 400 Hz (polyphase resampling), truncated to 4000 samples, and zero-padded symmetrically to 4096.

Duplicate waveforms. Two records were treated as duplicates when the MD5 hashes of their 250-Hz signals, quantised to 0.005 mV, were equal, or when their 40-ms block means differed by less than 0.01 mV in every lead.

S2. Models

In-house models. One-dimensional residual convolutional network: - four residual blocks with 32, 64, 128, and 256 channels and stride 2; - kernel sizes 7 and 5, group normalisation, and SiLU activation; - global average pooling and a linear head.

Inputs were divided by a fixed lead-wise RMS scale estimated in training patients. Training details: - 298 542 adult I0001 training patients with valid signals (one ECG each, consistent recorded sex); - AdamW (weight decay 10^{-4}), one-cycle learning rate (maximum 2×10^{-3}), 12 epochs, batch size 512, bfloat16 mixed precision; - the best epoch selected in 19 896 validation patients: lowest mean absolute error for age, highest AUC for sex.

Five seeds were trained for each task. Validation mean absolute error for age was 7.65–7.67 years; validation AUC for sex was 0.960–0.961. Secondary sex models using only the limb or only the precordial leads were trained earlier on 100 000 patients and were used without change.

Published models.

Model	Training data	Input	Choice made in I0001 validation patients	Validation result
Lima <i>et al.</i> 2021, ECG age	CODE (Brazil)	400 Hz, 4096 samples	amplitude $\times 1$ vs $\times 10$	$\times 1$: MAE 10.4 y, r 0.73; $\times 10$: MAE 14.7 y

Model	Training data	Input	Choice made in I0001 validation patients	Validation result
Bracke <i>et al.</i> 2026, ECG age	CODE-15% (Brazil)	400 Hz, 4096 samples, recorded sex	none	MAE 8.3 y, r 0.82 (published CODE-15%: 6.72 y, 0.9)
Ribeiro <i>et al.</i> 2020, abnormalities	CODE (Brazil)	400 Hz, 4096 samples	amplitude ×1 vs ×10	×1 mean AUC 0.983; ×10 0.968

Validation used 5000 I0001 validation patients (4981 with valid signals). The Bracke model was run with mamba-ssm installed without its CUDA extension, through the library's Triton-kernel path (`use_mem_eff_path=False`). This path computes the same function as the fused kernel. All weights loaded with no missing or unexpected parameters. Per-class validation AUCs of the Ribeiro model against GE 12SL v24 statements were First-degree AV block 0.986, RBBB 0.998, LBBB 0.999, Sinus bradycardia 0.930, Atrial fibrillation 0.988, Sinus tachycardia 0.996.

S3. Signal descriptors

- **Heart rate and RR-interval coefficient of variation:** from QRS detection on lead II (WFDB XQRS).
- **High-frequency and baseline power:** fractions of spectral power above 40 Hz and below 0.5 Hz; the median across leads.
- **Lead RMS:** root-mean-square amplitude per lead.
- **Amplitude change between ECGs:** the mean absolute log ratio of lead RMS, computed for leads I and II (limb) and V1–V6 (precordial).

S4. Electrode-reversal detection

For each pair, a median beat was formed for each ECG from beats detected on a multilead energy envelope (window −250 to +450 ms; baseline set to the first 40 ms). Limb-electrode permutations were applied to the second ECG through Einthoven's relations on leads I and II:

Permutation	Lead I'	Lead II'
RA ↔ LA	−I	II − I
LA ↔ LL	II	I
RA ↔ LL	I − II	−II
RA → LA → LL → RA	II − I	−I
RA → LL → LA → RA	−II	I − II

Each transformed beat was compared with the first ECG's beat after optimal scalar gain and a shift of up to ±20 ms. The ratio of the best permutation's relative residual to the identity residual was the limb score; a pair was flagged if the score was below 0.8. The same was done for the five adjacent precordial transpositions (V1 ↔ V2 ... V5 ↔ V6), with a threshold of 0.7. Thresholds were chosen against MIMIC machine statements and applied unchanged to HEEDB (*Table S8*).

S5. Agreement statistics

For paired values a_i, b_i in n patients: - within-patient SD: $S_w = SD(b - a)/\sqrt{2}$; - repeatability coefficient: $2.77 \cdot S_w (= 1.96 \cdot \sqrt{2} \cdot S_w)$; - limits of agreement: $\text{mean}(b - a) \pm 1.96 \cdot SD(b - a)$.

ICC(A,1) was computed from two-way ANOVA mean squares for rows (MSR), columns (MSC), and error (MSE):

$$\text{ICC}(A,1) = (\text{MSR} - \text{MSE}) / (\text{MSR} + \text{MSE} + 2(\text{MSC} - \text{MSE})/n).$$

Cohen's κ was computed for binary calls. All 95% confidence intervals are percentile intervals from 2000 bootstrap resamples of patients; each patient contributed one pair per stratum. For the age gap at the second ECG, chronological age was advanced by the interval between ECGs.

S6. Protocol and deviations

The analysis protocol was written before model outputs were computed and is kept with the code together with a dated change log.

Changes made before results were seen: - implementation of duplicate detection; - the 18–89-year age window; - in-house model training details; - the published-model amplitude multiplier; - the three-ECG analysis.

Analyses added after the primary results (post hoc): - signal change by interval; - age-gap variability by quintile of precordial amplitude change; - signal-similar 1–6-hour pairs; - sex outputs restricted to adults; - artifact and electrode-reversal analyses; - the age-bias-corrected gap.

Models added after the primary results: the second published ECG-age model and the abnormality classifier, added to test whether the findings depend on a single published model. Their analyses were specified before their outputs were computed.

Supplementary Tables

Table S1. Pair flow by cohort and interval.

Cohort	Interval	Pairs sampled	Both ECGs valid	Duplicates removed	Age analyses	Sex analyses	Median interval (min)
I0001	<5 min	25 000	24 004	0	19 106	23 990	0.9
I0001	5–60 min	24 626	23 888	0	17 393	23 838	23.0
I0001	1–6 h	25 000	24 525	0	17 409	24 467	190.2
I0001	6–24 h	25 000	23 974	0	15 194	23 919	819.6
I0001	1–7 d	25 000	23 737	0	14 426	23 680	3161.9
I0001	7–30 d	25 000	24 014	0	15 785	23 959	21631.0
I0001	30 d–1 y	25 000	24 017	1	15 792	23 966	185650.2
I0001	>1 y	25 000	23 741	0	13 697	23 682	1560431.8
I0006	<5 min	17 310	16 438	0	15 513	16 423	0.8
I0006	5–60 min	10 011	9 802	0	9 401	9 788	19.4
I0006	1–6 h	22 762	22 623	0	21 507	22 605	212.0
I0006	6–24 h	25 000	24 894	0	23 446	24 881	912.3
I0006	1–7 d	25 000	24 892	0	23 505	24 839	3494.7
I0006	7–30 d	25 000	24 907	0	23 636	24 723	21491.2
I0006	30 d–1 y	25 000	24 909	0	23 507	24 151	221029.8
I0006	>1 y	25 000	24 915	0	23 618	24 054	940127.6
MIMIC	<5 min	7 416	6 548	0	6 026	6 546	1.0
MIMIC	5–60 min	12 498	11 832	0	10 848	11 831	24.0
MIMIC	1–6 h	25 000	24 124	15	22 532	24 107	199.0
MIMIC	6–24 h	25 000	24 114	0	22 413	24 109	717.0
MIMIC	1–7 d	25 000	24 018	0	22 236	24 016	3431.0
MIMIC	7–30 d	25 000	24 231	0	22 831	24 230	21533.0
MIMIC	30 d–1 y	25 000	24 400	0	23 087	24 398	175556.5
MIMIC	>1 y	25 000	24 454	0	23 675	24 452	1100431.5

Cohort	Interval	Pairs sampled	Both ECGs valid	Duplicates removed	Age analyses	Sex analyses	Median interval (min)
MIMIC ED	1–6 h	3 911	3 911	5	3 780	3 906	280.0

Table S2a. Age-gap agreement, Lima *et al.* model.

Cohort	Interval	n	S_w , y (95% CI)	RC, y	Bias (LoA), y	ICC age gap (95% CI)	ICC predicted age	Class change >8 y, % (95% CI)	κ	MAE first ECG, y
I0001	<5 min	19 106	3.74 (3.66–3.82)	10.4	-0.3 (-10.7, 10.1)	0.92 (0.92–0.92)	0.94	10.8 (10.3–11.2)	0.76	10.7
I0001	5–60 min	17 393	6.19 (6.09–6.30)	17.2	-0.6 (-17.8, 16.5)	0.80 (0.79–0.80)	0.83	17.7 (17.1–18.3)	0.63	11.3
I0001	1–6 h	17 409	6.52 (6.41–6.64)	18.1	-0.7 (-18.8, 17.4)	0.77 (0.76–0.77)	0.82	18.6 (18.1–19.2)	0.60	11.0
I0001	6–24 h	15 194	7.07 (6.96–7.18)	19.6	-0.7 (-20.3, 18.9)	0.74 (0.73–0.74)	0.78	20.5 (19.8–21.2)	0.55	11.1
I0001	1–7 d	14 426	7.77 (7.64–7.89)	21.5	-0.2 (-21.7, 21.4)	0.68 (0.67–0.69)	0.73	22.6 (21.9–23.3)	0.50	11.0
I0001	7–30 d	15 785	7.55 (7.44–7.65)	20.9	0.3 (-20.6, 21.2)	0.66 (0.65–0.67)	0.74	22.7 (22.0–23.3)	0.49	10.3
I0001	30 d–1 y	15 792	7.00 (6.89–7.10)	19.4	0.2 (-19.2, 19.6)	0.68 (0.67–0.69)	0.79	21.5 (20.8–22.1)	0.51	10.0
I0001	>1 y	13 697	7.52 (7.39–7.65)	20.8	-0.9 (-21.8, 19.9)	0.60 (0.59–0.62)	0.73	24.2 (23.5–24.9)	0.44	9.7
I0006	<5 min	15 513	4.06 (3.98–4.15)	11.3	-0.1 (-11.4, 11.1)	0.91 (0.90–0.91)	0.92	12.3 (11.8–12.8)	0.74	10.9
I0006	5–60 min	9 401	7.19 (7.03–7.35)	19.9	0.7 (-19.2, 20.6)	0.74 (0.73–0.75)	0.75	20.5 (19.7–21.3)	0.58	11.4
I0006	1–6 h	21 507	6.07 (5.98–6.15)	16.8	-0.4 (-17.2, 16.4)	0.80 (0.79–0.81)	0.79	19.2 (18.6–19.7)	0.61	11.8
I0006	6–24 h	23 446	6.47 (6.39–6.55)	17.9	0.1 (-17.8, 18.0)	0.78 (0.77–0.78)	0.75	20.4 (19.9–20.9)	0.59	12.0
I0006	1–7 d	23 505	7.19 (7.10–7.27)	19.9	0.3 (-19.6, 20.2)	0.73 (0.73–0.74)	0.72	22.1 (21.6–22.7)	0.55	11.9
I0006	7–30 d	23 636	7.25 (7.16–7.33)	20.1	-0.1 (-20.2, 20.0)	0.72 (0.71–0.73)	0.74	22.4 (21.9–23.0)	0.54	11.4
I0006	30 d–1 y	23 507	7.47 (7.36–7.57)	20.7	-0.0 (-20.7, 20.7)	0.70 (0.69–0.71)	0.73	22.0 (21.5–22.6)	0.53	10.8
I0006	>1 y	23 618	7.36 (7.26–7.46)	20.4	-0.5 (-20.9, 19.9)	0.68 (0.67–0.69)	0.74	21.1 (20.5–21.6)	0.52	10.1
MIMIC	<5 min	6 026	4.25 (4.12–4.38)	11.8	-0.4 (-12.2, 11.4)	0.88 (0.88–0.89)	0.90	12.3 (11.5–13.1)	0.73	10.3
MIMIC	5–60 min	10 848	5.57 (5.43–5.71)	15.4	-0.6 (-16.1, 14.8)	0.81 (0.80–0.82)	0.85	16.3 (15.6–17.0)	0.66	10.8
MIMIC	1–6 h	22 532	5.39 (5.32–5.47)	14.9	-0.5 (-15.4, 14.5)	0.81 (0.80–0.81)	0.86	17.5 (17.1–18.0)	0.63	10.3
MIMIC	6–24 h	22 413	5.59 (5.51–5.66)	15.5	-0.3 (-15.8, 15.2)	0.79 (0.78–0.79)	0.85	17.7 (17.2–18.2)	0.61	10.0
MIMIC	1–7 d	22 236	6.21 (6.14–6.29)	17.2	-0.4 (-17.6, 16.9)	0.75 (0.74–0.75)	0.81	19.8 (19.3–20.4)	0.57	10.1
MIMIC	7–30 d	22 831	6.22 (6.15–6.30)	17.2	0.2 (-17.1, 17.4)	0.74 (0.73–0.75)	0.81	19.9 (19.4–20.5)	0.56	10.0
MIMIC	30 d–1 y	23 087	5.92 (5.85–5.99)	16.4	0.1 (-16.3, 16.6)	0.75 (0.74–0.75)	0.84	19.3 (18.8–19.8)	0.56	9.5
MIMIC	>1 y	23 675	6.08 (6.02–6.16)	16.9	-0.3 (-17.1, 16.6)	0.72 (0.71–0.73)	0.82	20.1 (19.6–20.6)	0.53	9.2
MIMIC ED	1–6 h	3 780	4.73 (4.58–4.89)	13.1	-0.5 (-13.6, 12.6)	0.83 (0.82–0.84)	0.89	16.8 (15.7–18.0)	0.63	9.8

Table S2b. Age-gap agreement, Bracke *et al.* model. ECGs with unknown or conflicting recorded sex were not scored by this model.

Cohort	Interval	n	S_w , y (95% CI)	RC, y	Bias (LoA), y	ICC age gap (95% CI)	ICC predicted age	Class change >8 y, % (95% CI)	κ	MAE first ECG, y
I0001	<5 min	19 104	4.16 (4.09–4.24)	11.5	-0.2 (-11.7, 11.4)	0.87 (0.87–0.88)	0.94	10.7 (10.2–11.1)	0.69	9.0
I0001	5–60 min	17 393	6.47 (6.37–6.58)	17.9	-0.3 (-18.3, 17.6)	0.74 (0.73–0.75)	0.84	16.5 (16.0–17.1)	0.56	9.7
I0001	1–6 h	17 408	6.79 (6.69–6.90)	18.8	-0.4 (-19.2, 18.5)	0.69 (0.68–0.70)	0.83	17.5 (16.9–18.1)	0.50	9.4
I0001	6–24 h	15 193	7.22 (7.11–7.34)	20.0	-0.7 (-20.7, 19.4)	0.64 (0.63–0.66)	0.79	18.2 (17.6–18.9)	0.45	9.3
I0001	1–7 d	14 425	7.66 (7.54–7.78)	21.2	-0.1 (-21.3, 21.1)	0.59 (0.58–0.60)	0.77	19.6 (19.0–20.3)	0.41	9.1
I0001	7–30 d	15 782	7.34 (7.23–7.45)	20.3	0.2 (-20.1, 20.6)	0.57 (0.56–0.59)	0.78	19.7 (19.1–20.4)	0.41	8.5

Cohort	Interval	n	S_w , y (95% CI)	RC, y	Bias (LoA), y	ICC age gap (95% CI)	ICC predicted age	Class change >8 y, % (95% CI)	κ	MAE first ECG, y
I0001	30 d–1 y	15 790	6.62 (6.52–6.72)	18.3	0.3 (-18.0, 18.7)	0.59 (0.58–0.60)	0.82	18.0 (17.4–18.6)	0.41	8.0
I0001	>1 y	13 697	6.81 (6.70–6.93)	18.9	-0.3 (-19.2, 18.6)	0.53 (0.51–0.54)	0.76	18.5 (17.8–19.2)	0.36	7.5
I0006	<5 min	15 503	4.17 (4.09–4.26)	11.6	-0.3 (-11.8, 11.3)	0.87 (0.87–0.88)	0.93	11.0 (10.6–11.5)	0.68	9.0
I0006	5–60 min	9 393	7.17 (7.01–7.32)	19.9	-0.4 (-20.3, 19.4)	0.68 (0.66–0.69)	0.78	15.6 (14.8–16.3)	0.53	9.5
I0006	1–6 h	21 498	6.53 (6.43–6.62)	18.1	0.0 (-18.1, 18.1)	0.72 (0.71–0.73)	0.79	18.3 (17.8–18.8)	0.55	9.5
I0006	6–24 h	23 442	7.00 (6.92–7.09)	19.4	-0.1 (-19.5, 19.4)	0.69 (0.68–0.70)	0.77	20.6 (20.0–21.1)	0.51	9.7
I0006	1–7 d	23 498	7.56 (7.47–7.65)	20.9	0.0 (-20.9, 20.9)	0.65 (0.64–0.65)	0.74	21.5 (20.9–22.0)	0.48	9.7
I0006	7–30 d	23 623	7.47 (7.38–7.56)	20.7	-0.2 (-20.9, 20.5)	0.63 (0.62–0.64)	0.75	20.0 (19.5–20.5)	0.49	9.2
I0006	30 d–1 y	23 484	7.50 (7.40–7.61)	20.8	0.0 (-20.8, 20.8)	0.60 (0.59–0.61)	0.74	20.2 (19.7–20.7)	0.47	8.7
I0006	>1 y	23 597	7.17 (7.07–7.27)	19.9	-0.2 (-20.1, 19.6)	0.58 (0.57–0.59)	0.73	18.4 (17.9–18.8)	0.44	8.1
MIMIC	<5 min	6 026	4.54 (4.41–4.69)	12.6	-0.4 (-13.0, 12.2)	0.84 (0.83–0.85)	0.91	12.3 (11.5–13.2)	0.70	9.0
MIMIC	5–60 min	10 848	6.02 (5.89–6.16)	16.7	-0.4 (-17.1, 16.3)	0.73 (0.72–0.75)	0.85	17.5 (16.8–18.2)	0.58	9.2
MIMIC	1–6 h	22 532	5.84 (5.76–5.92)	16.2	-0.2 (-16.4, 15.9)	0.72 (0.72–0.73)	0.86	18.1 (17.6–18.6)	0.55	8.8
MIMIC	6–24 h	22 413	6.02 (5.94–6.10)	16.7	-0.4 (-17.1, 16.2)	0.70 (0.69–0.71)	0.85	18.5 (18.0–19.0)	0.51	8.6
MIMIC	1–7 d	22 236	6.77 (6.68–6.86)	18.8	-0.5 (-19.3, 18.3)	0.65 (0.64–0.66)	0.82	20.0 (19.4–20.5)	0.48	8.9
MIMIC	7–30 d	22 831	6.62 (6.54–6.70)	18.3	-0.1 (-18.4, 18.3)	0.65 (0.64–0.65)	0.81	19.7 (19.2–20.3)	0.48	8.6
MIMIC	30 d–1 y	23 087	6.14 (6.06–6.22)	17.0	0.3 (-16.7, 17.3)	0.65 (0.64–0.66)	0.85	19.5 (19.0–20.0)	0.46	8.1
MIMIC	>1 y	23 675	6.23 (6.16–6.31)	17.3	0.0 (-17.2, 17.3)	0.62 (0.61–0.63)	0.82	19.4 (18.9–19.9)	0.42	7.7
MIMIC ED	1–6 h	3 780	4.85 (4.70–4.99)	13.4	-0.2 (-13.6, 13.3)	0.76 (0.75–0.78)	0.87	16.0 (14.9–17.2)	0.55	7.9

Table S2c. Age-gap agreement, in-house five-seed ensemble.

Cohort	Interval	n	S_w , y (95% CI)	RC, y	Bias (LoA), y	ICC age gap (95% CI)	ICC predicted age	Class change >8 y, % (95% CI)	κ	MAE first ECG, y
I0001	<5 min	19 106	2.95 (2.88–3.01)	8.2	-0.2 (-8.4, 8.0)	0.91 (0.91–0.92)	0.96	8.3 (8.0–8.8)	0.76	7.8
I0001	5–60 min	17 393	4.80 (4.70–4.90)	13.3	-0.4 (-13.7, 13.0)	0.80 (0.79–0.81)	0.89	14.0 (13.4–14.5)	0.63	8.3
I0001	1–6 h	17 409	4.89 (4.79–4.99)	13.5	-0.1 (-13.7, 13.4)	0.78 (0.77–0.78)	0.89	14.1 (13.6–14.6)	0.61	7.9
I0001	6–24 h	15 194	5.05 (4.94–5.15)	14.0	-0.3 (-14.2, 13.7)	0.76 (0.75–0.77)	0.88	14.6 (14.1–15.1)	0.58	7.9
I0001	1–7 d	14 426	5.47 (5.37–5.57)	15.2	-0.2 (-15.3, 15.0)	0.72 (0.71–0.73)	0.86	16.0 (15.5–16.7)	0.53	7.8
I0001	7–30 d	15 785	5.35 (5.25–5.45)	14.8	-0.1 (-14.9, 14.8)	0.70 (0.69–0.71)	0.86	16.7 (16.1–17.2)	0.50	7.4
I0001	30 d–1 y	15 792	4.91 (4.82–5.00)	13.6	0.1 (-13.5, 13.7)	0.71 (0.70–0.72)	0.89	15.2 (14.6–15.7)	0.52	7.1
I0001	>1 y	13 697	5.42 (5.30–5.54)	15.0	-0.7 (-15.7, 14.3)	0.63 (0.61–0.64)	0.83	17.4 (16.8–18.1)	0.43	6.8
I0006	<5 min	15 513	3.02 (2.96–3.09)	8.4	-0.1 (-8.5, 8.3)	0.91 (0.91–0.92)	0.95	9.2 (8.7–9.6)	0.75	8.0
I0006	5–60 min	9 401	5.45 (5.30–5.59)	15.1	-0.6 (-15.7, 14.6)	0.75 (0.74–0.76)	0.83	14.4 (13.7–15.2)	0.60	8.2
I0006	1–6 h	21 507	4.59 (4.52–4.67)	12.7	-0.2 (-13.0, 12.5)	0.82 (0.81–0.82)	0.87	14.3 (13.8–14.8)	0.64	8.5
I0006	6–24 h	23 446	4.85 (4.79–4.92)	13.4	-0.1 (-13.5, 13.4)	0.81 (0.80–0.81)	0.85	15.2 (14.7–15.6)	0.63	8.6
I0006	1–7 d	23 505	5.29 (5.23–5.36)	14.7	0.0 (-14.6, 14.7)	0.77 (0.76–0.78)	0.84	16.4 (15.9–16.9)	0.59	8.5
I0006	7–30 d	23 636	5.27 (5.19–5.34)	14.6	-0.1 (-14.7, 14.5)	0.76 (0.75–0.77)	0.84	15.8 (15.3–16.2)	0.59	8.2
I0006	30 d–1 y	23 507	5.73 (5.63–5.83)	15.9	-0.1 (-15.9, 15.8)	0.71 (0.70–0.72)	0.82	15.7 (15.2–16.1)	0.57	7.8
I0006	>1 y	23 618	5.63 (5.53–5.73)	15.6	-0.4 (-16.0, 15.2)	0.68 (0.67–0.69)	0.80	14.8 (14.4–15.3)	0.54	7.2

Cohort	Interval	n	S_w , y (95% CI)	RC, y	Bias (LoA), y	ICC age gap (95% CI)	ICC predicted age	Class change >8 y, % (95% CI)	κ	MAE first ECG, y
MIMIC	<5 min	6 026	3.19 (3.07–3.31)	8.8	-0.1 (-9.0, 8.7)	0.90 (0.89–0.90)	0.94	9.3 (8.6–10.1)	0.74	7.8
MIMIC	5–60 min	10 848	4.47 (4.33–4.61)	12.4	-0.3 (-12.7, 12.1)	0.81 (0.80–0.82)	0.91	13.0 (12.4–13.7)	0.65	8.0
MIMIC	1–6 h	22 532	4.11 (4.05–4.18)	11.4	-0.2 (-11.6, 11.1)	0.82 (0.81–0.82)	0.92	13.5 (13.1–14.0)	0.62	7.6
MIMIC	6–24 h	22 413	4.22 (4.16–4.28)	11.7	-0.3 (-12.0, 11.4)	0.80 (0.80–0.81)	0.91	13.6 (13.2–14.0)	0.60	7.4
MIMIC	1–7 d	22 236	4.69 (4.62–4.76)	13.0	-0.4 (-13.4, 12.6)	0.77 (0.76–0.78)	0.89	14.5 (14.1–15.0)	0.57	7.6
MIMIC	7–30 d	22 831	4.69 (4.63–4.76)	13.0	-0.1 (-13.1, 13.0)	0.76 (0.75–0.77)	0.89	14.8 (14.3–15.3)	0.56	7.4
MIMIC	30 d–1 y	23 087	4.49 (4.43–4.56)	12.4	0.2 (-12.3, 12.6)	0.76 (0.75–0.77)	0.90	14.8 (14.4–15.3)	0.54	7.1
MIMIC	>1 y	23 675	4.66 (4.60–4.71)	12.9	-0.0 (-12.9, 12.9)	0.73 (0.72–0.73)	0.87	15.1 (14.7–15.6)	0.50	6.8
MIMIC ED	1–6 h	3 780	3.67 (3.54–3.81)	10.2	-0.3 (-10.4, 9.9)	0.83 (0.82–0.84)	0.92	13.4 (12.3–14.5)	0.63	7.3

Table S2d. Age-gap agreement, in-house single seed (seed 1).

Cohort	Interval	n	S_w , y (95% CI)	RC, y	Bias (LoA), y	ICC age gap (95% CI)	ICC predicted age	Class change >8 y, % (95% CI)	κ	MAE first ECG, y
I0001	<5 min	19 106	3.57 (3.50–3.64)	9.9	-0.2 (-10.1, 9.7)	0.88 (0.88–0.89)	0.95	10.5 (10.0–10.9)	0.70	8.1
I0001	5–60 min	17 393	5.42 (5.31–5.52)	15.0	-0.4 (-15.4, 14.6)	0.76 (0.75–0.77)	0.87	15.3 (14.8–15.9)	0.60	8.6
I0001	1–6 h	17 409	5.44 (5.33–5.55)	15.1	-0.2 (-15.2, 14.9)	0.74 (0.73–0.75)	0.87	15.4 (14.8–15.9)	0.58	8.1
I0001	6–24 h	15 194	5.65 (5.55–5.76)	15.7	-0.2 (-15.9, 15.4)	0.72 (0.71–0.73)	0.85	16.5 (15.9–17.1)	0.53	8.1
I0001	1–7 d	14 426	5.98 (5.88–6.09)	16.6	-0.1 (-16.7, 16.4)	0.68 (0.67–0.70)	0.84	17.3 (16.7–17.9)	0.51	8.1
I0001	7–30 d	15 785	5.85 (5.75–5.95)	16.2	-0.1 (-16.3, 16.1)	0.66 (0.65–0.67)	0.84	18.1 (17.5–18.7)	0.47	7.6
I0001	30 d–1 y	15 792	5.33 (5.24–5.42)	14.8	0.1 (-14.6, 14.9)	0.68 (0.67–0.69)	0.87	16.9 (16.3–17.4)	0.49	7.2
I0001	>1 y	13 697	5.82 (5.70–5.93)	16.1	-0.7 (-16.8, 15.5)	0.59 (0.58–0.61)	0.81	18.4 (17.7–19.0)	0.41	7.0
I0006	<5 min	15 513	3.66 (3.59–3.75)	10.2	0.0 (-10.1, 10.2)	0.88 (0.88–0.89)	0.93	10.7 (10.2–11.2)	0.71	8.3
I0006	5–60 min	9 401	6.06 (5.91–6.20)	16.8	-0.6 (-17.4, 16.2)	0.72 (0.70–0.73)	0.81	16.0 (15.3–16.8)	0.55	8.5
I0006	1–6 h	21 507	5.24 (5.16–5.32)	14.5	-0.2 (-14.7, 14.3)	0.78 (0.77–0.79)	0.84	15.7 (15.2–16.1)	0.61	8.7
I0006	6–24 h	23 446	5.50 (5.43–5.58)	15.2	-0.0 (-15.3, 15.2)	0.77 (0.76–0.78)	0.82	16.7 (16.2–17.1)	0.59	8.9
I0006	1–7 d	23 505	5.92 (5.84–6.00)	16.4	0.1 (-16.4, 16.5)	0.73 (0.73–0.74)	0.81	18.1 (17.6–18.6)	0.55	8.8
I0006	7–30 d	23 636	5.89 (5.81–5.97)	16.3	-0.2 (-16.5, 16.1)	0.72 (0.71–0.73)	0.82	17.3 (16.8–17.8)	0.55	8.4
I0006	30 d–1 y	23 507	6.24 (6.13–6.34)	17.3	-0.1 (-17.4, 17.2)	0.68 (0.67–0.69)	0.80	16.9 (16.4–17.4)	0.54	8.0
I0006	>1 y	23 618	6.10 (6.00–6.20)	16.9	-0.4 (-17.3, 16.5)	0.65 (0.64–0.66)	0.78	15.9 (15.4–16.4)	0.51	7.4
MIMIC	<5 min	6 026	3.76 (3.63–3.90)	10.4	-0.1 (-10.5, 10.4)	0.87 (0.86–0.88)	0.92	11.1 (10.3–11.9)	0.70	8.0
MIMIC	5–60 min	10 848	5.00 (4.86–5.14)	13.8	-0.3 (-14.1, 13.6)	0.78 (0.76–0.79)	0.89	14.1 (13.4–14.7)	0.62	8.3
MIMIC	1–6 h	22 532	4.59 (4.52–4.66)	12.7	-0.3 (-13.0, 12.5)	0.78 (0.78–0.79)	0.90	15.0 (14.5–15.5)	0.59	7.8
MIMIC	6–24 h	22 413	4.69 (4.62–4.76)	13.0	-0.3 (-13.3, 12.7)	0.77 (0.76–0.78)	0.89	14.8 (14.3–15.2)	0.58	7.7
MIMIC	1–7 d	22 236	5.16 (5.09–5.23)	14.3	-0.5 (-14.8, 13.9)	0.73 (0.73–0.74)	0.87	16.0 (15.5–16.4)	0.54	7.8
MIMIC	7–30 d	22 831	5.13 (5.06–5.19)	14.2	-0.0 (-14.3, 14.2)	0.73 (0.72–0.73)	0.87	16.4 (16.0–16.9)	0.52	7.6
MIMIC	30 d–1 y	23 087	4.89 (4.83–4.95)	13.6	0.2 (-13.3, 13.8)	0.73 (0.72–0.74)	0.89	16.3 (15.8–16.8)	0.51	7.3
MIMIC	>1 y	23 675	5.03 (4.97–5.09)	13.9	-0.0 (-13.9, 13.9)	0.70 (0.69–0.70)	0.86	16.2 (15.7–16.7)	0.48	7.0
MIMIC ED	1–6 h	3 780	4.11 (3.97–4.24)	11.4	-0.2 (-11.6, 11.1)	0.80 (0.79–0.81)	0.91	14.8 (13.7–15.9)	0.60	7.5

Table S3. Sex-model agreement. Limb-only and precordial-only models have their own logit scales and are not directly comparable with the 12-lead ensemble.

Cohort	Interval	n	Logit S_w (95% CI)	ICC all	ICC female	ICC male	Class change, % (95% CI)	Class change single seed, %	Abs. change in $p > 0.2$, %	Limb-only S_w	Precordial-only S_w
I0001	<5 min	23 990	0.88 (0.86–0.89)	0.94	0.89	0.90	8.6 (8.2–9.0)	10.9	12.4	0.35	0.40
I0001	5–60 min	23 838	1.28 (1.25–1.30)	0.85	0.75	0.76	14.4 (13.9–14.9)	17.0	22.8	0.50	0.63
I0001	1–6 h	24 467	1.31 (1.29–1.33)	0.86	0.77	0.74	14.3 (13.9–14.8)	16.8	24.2	0.52	0.67
I0001	6–24 h	23 919	1.39 (1.37–1.40)	0.84	0.74	0.71	15.4 (14.9–15.8)	17.9	25.7	0.54	0.72
I0001	1–7 d	23 680	1.54 (1.53–1.56)	0.80	0.67	0.65	17.7 (17.2–18.2)	20.0	30.1	0.57	0.79
I0001	7–30 d	23 959	1.63 (1.62–1.65)	0.81	0.65	0.65	16.8 (16.3–17.2)	18.6	29.0	0.58	0.83
I0001	30 d–1 y	23 966	1.68 (1.66–1.70)	0.84	0.67	0.67	14.0 (13.6–14.5)	15.9	24.8	0.59	0.82
I0001	>1 y	23 682	1.93 (1.91–1.95)	0.82	0.60	0.63	13.7 (13.3–14.1)	15.3	24.8	0.67	0.90
I0006	<5 min	16 423	0.91 (0.89–0.92)	0.94	0.87	0.90	8.2 (7.8–8.6)	10.3	12.3	0.39	0.43
I0006	5–60 min	9 788	1.51 (1.48–1.55)	0.82	0.67	0.69	15.9 (15.2–16.7)	18.1	26.6	0.61	0.76
I0006	1–6 h	22 605	1.33 (1.32–1.35)	0.84	0.70	0.72	15.3 (14.8–15.7)	17.5	25.3	0.51	0.73
I0006	6–24 h	24 881	1.40 (1.38–1.41)	0.80	0.63	0.67	17.5 (17.0–18.0)	19.7	29.1	0.53	0.78
I0006	1–7 d	24 839	1.54 (1.52–1.55)	0.77	0.59	0.60	18.8 (18.3–19.2)	21.0	31.9	0.58	0.85
I0006	7–30 d	24 723	1.60 (1.58–1.62)	0.80	0.60	0.60	17.6 (17.1–18.1)	19.4	29.9	0.60	0.87
I0006	30 d–1 y	24 151	1.62 (1.60–1.64)	0.83	0.61	0.64	15.6 (15.1–16.0)	17.3	27.1	0.61	0.87
I0006	>1 y	24 054	1.67 (1.66–1.69)	0.85	0.62	0.62	13.1 (12.7–13.5)	14.7	23.7	0.62	0.88
MIMIC	<5 min	6 546	0.94 (0.91–0.97)	0.92	0.86	0.85	10.4 (9.6–11.1)	12.4	15.3	0.39	0.46
MIMIC	5–60 min	11 831	1.23 (1.20–1.26)	0.87	0.75	0.77	13.4 (12.8–14.0)	15.8	22.0	0.48	0.64
MIMIC	1–6 h	24 107	1.30 (1.28–1.31)	0.88	0.74	0.75	13.0 (12.6–13.4)	15.3	22.6	0.51	0.68
MIMIC	6–24 h	24 109	1.36 (1.35–1.38)	0.87	0.72	0.73	14.2 (13.7–14.6)	16.0	24.3	0.54	0.73
MIMIC	1–7 d	24 016	1.50 (1.49–1.52)	0.81	0.62	0.64	16.7 (16.3–17.2)	18.7	29.0	0.58	0.82
MIMIC	7–30 d	24 230	1.58 (1.56–1.60)	0.82	0.61	0.63	15.8 (15.4–16.3)	17.6	27.8	0.60	0.85
MIMIC	30 d–1 y	24 398	1.61 (1.60–1.63)	0.84	0.63	0.64	14.3 (13.9–14.8)	15.6	25.2	0.61	0.86

Cohort	Interval	n	Logit S_w (95% CI)	ICC all	ICC female	ICC male	Class change, % (95% CI)	Class change single seed, %	Abs. change in $p > 0.2$, %	Limb-only S_w	Precordial-only S_w
MIMIC	>1 y	24 452	1.71 (1.70–1.73)	0.85	0.60	0.60	13.1 (12.6–13.5)	14.4	23.9	0.67	0.90
MIMIC ED	1–6 h	3 906	1.30 (1.27–1.33)	0.92	0.74	0.79	8.7 (7.8–9.6)	10.4	15.7	0.53	0.63

Table S4. Variance of the age gap at 1–6 hours as a percentage of the variance at >1 year.

Cohort	Lima	Bracke	In-house ensemble
I0001	75	99	81
I0006	68	83	67
MIMIC	79	88	78

Table S5. Age-bias-corrected age gap at 1–6 hours (post hoc). The gap was replaced by its residual from a site- and model-specific linear regression on chronological age.

Cohort	Model	Slope of gap on age	S_w , y	ICC	Prevalence >8 y, %	Class change, %	κ
I0001	Lima <i>et al.</i>	-0.368	6.52	0.71	29.1	19.1	0.53
I0001	Bracke <i>et al.</i>	-0.236	6.79	0.65	25.2	19.2	0.48
I0001	In-house five-seed ensemble	-0.240	4.89	0.74	19.9	13.6	0.57
I0006	Lima <i>et al.</i>	-0.454	6.07	0.72	29.3	19.6	0.52
I0006	Bracke <i>et al.</i>	-0.319	6.53	0.66	26.1	20.2	0.48
I0006	In-house five-seed ensemble	-0.351	4.59	0.75	20.9	14.3	0.56
MIMIC	Lima <i>et al.</i>	-0.364	5.39	0.74	25.3	17.5	0.53
MIMIC	Bracke <i>et al.</i>	-0.247	5.84	0.67	24.0	17.7	0.51
MIMIC	In-house five-seed ensemble	-0.264	4.11	0.77	18.7	13.0	0.56
MIMIC ED	Lima <i>et al.</i>	-0.329	4.73	0.80	23.7	14.9	0.57
MIMIC ED	Bracke <i>et al.</i>	-0.281	4.85	0.72	18.6	15.2	0.49
MIMIC ED	In-house five-seed ensemble	-0.267	3.67	0.80	15.9	10.6	0.59

Table S6. Correlates of absolute change at 1–6 hours. Ordinary least squares with HC3 standard errors; coefficients are per SD of continuous predictors or per unit of binary predictors, with 95% CIs.

Table S6a. HEEDB I0001 (test), 1–6 h.

Predictor	Lima: abs. change in age gap, y	Bracke: abs. change in age gap, y	In-house: abs. change in age gap, y	Sex: abs. change in logit
Heart-rate difference	+1.10 (+0.95, +1.24)	+1.07 (+0.93, +1.21)	+0.78 (+0.67, +0.89)	+0.11 (+0.08, +0.13)
RR variability (higher of pair)	-0.29 (-0.40, -0.19)	+0.05 (-0.07, +0.16)	-0.16 (-0.25, -0.07)	-0.07 (-0.09, -0.05)
High-frequency noise (higher of pair, log)	+0.24 (+0.15, +0.34)	+0.40 (+0.29, +0.50)	+0.17 (+0.10, +0.24)	+0.02 (+0.00, +0.04)
Baseline wander (higher of pair,	-0.26 (-0.37, -0.16)	+0.00 (-0.11, +0.12)	-0.06 (-0.14, +0.03)	+0.00 (-0.02, +0.02)

Predictor	Lima: abs. change in age gap, y	Bracke: abs. change in age gap, y	In-house: abs. change in age gap, y	Sex: abs. change in logit
log)				
Limb amplitude change	+0.54 (+0.40, +0.69)	+0.35 (+0.20, +0.49)	+0.43 (+0.32, +0.55)	+0.05 (+0.02, +0.08)
Precordial amplitude change	+1.22 (+1.08, +1.37)	+0.57 (+0.44, +0.70)	+0.70 (+0.59, +0.81)	+0.21 (+0.18, +0.23)
Interval (log minutes)	+0.13 (+0.04, +0.22)	+0.12 (+0.02, +0.22)	+0.09 (+0.01, +0.16)	+0.02 (+0.00, +0.04)
Chronological age	-0.49 (-0.60, -0.39)	-0.04 (-0.14, +0.07)	-0.51 (-0.59, -0.42)	-0.08 (-0.10, -0.06)
Male (binary)	+0.01 (-0.17, +0.20)	-0.15 (-0.34, +0.05)	-0.32 (-0.47, -0.18)	-0.09 (-0.13, -0.05)
<i>n</i> ; <i>R</i> ²	17 394; 0.102	17 394; 0.055	17 394; 0.084	17 394; 0.054
Mean outcome	6.61	6.94	4.77	1.39

Table S6b. HEEDB I0006, 1–6 h.

Predictor	Lima: abs. change in age gap, y	Bracke: abs. change in age gap, y	In-house: abs. change in age gap, y	Sex: abs. change in logit
Heart-rate difference	+0.88 (+0.77, +0.99)	+0.88 (+0.77, +1.00)	+0.65 (+0.57, +0.73)	+0.08 (+0.06, +0.10)
RR variability (higher of pair)	-0.29 (-0.37, -0.21)	-0.28 (-0.38, -0.19)	-0.16 (-0.23, -0.09)	-0.09 (-0.11, -0.08)
High-frequency noise (higher of pair, log)	+0.20 (+0.12, +0.27)	+0.32 (+0.24, +0.40)	+0.13 (+0.07, +0.19)	+0.02 (-0.00, +0.03)
Baseline wander (higher of pair, log)	-0.22 (-0.30, -0.14)	-0.03 (-0.11, +0.06)	-0.09 (-0.15, -0.02)	-0.00 (-0.02, +0.01)
Limb amplitude change	+0.46 (+0.36, +0.57)	+0.35 (+0.25, +0.45)	+0.38 (+0.30, +0.46)	+0.05 (+0.03, +0.08)
Precordial amplitude change	+0.73 (+0.63, +0.83)	+0.72 (+0.61, +0.82)	+0.59 (+0.52, +0.67)	+0.21 (+0.18, +0.23)
Interval (log minutes)	-0.01 (-0.09, +0.07)	-0.03 (-0.11, +0.05)	-0.05 (-0.11, +0.02)	-0.00 (-0.02, +0.01)
Chronological age	-0.53 (-0.62, -0.45)	-0.36 (-0.45, -0.27)	-0.58 (-0.65, -0.51)	-0.05 (-0.07, -0.04)
Male (binary)	-0.19 (-0.35, -0.04)	-0.36 (-0.53, -0.19)	-0.43 (-0.55, -0.31)	-0.09 (-0.13, -0.06)
<i>n</i> ; <i>R</i> ²	21 472; 0.070	21 469; 0.056	21 472; 0.081	21 472; 0.048
Mean outcome	6.34	6.75	4.65	1.42

Table S6c. MIMIC-IV-ECG, 1–6 h.

Predictor	Lima: abs. change in age gap, y	Bracke: abs. change in age gap, y	In-house: abs. change in age gap, y	Sex: abs. change in logit
Heart-rate difference	+0.75 (+0.65, +0.85)	+0.67 (+0.56, +0.77)	+0.43 (+0.36, +0.51)	+0.08 (+0.06, +0.10)
RR variability (higher of pair)	-0.35 (-0.43, -0.27)	-0.25 (-0.33, -0.16)	-0.29 (-0.35, -0.23)	-0.08 (-0.10, -0.06)
High-frequency noise (higher of pair, log)	+0.14 (+0.08, +0.21)	+0.20 (+0.13, +0.27)	+0.08 (+0.03, +0.13)	+0.03 (+0.01, +0.04)
Baseline wander (higher of pair, log)	-0.03 (-0.10, +0.04)	+0.15 (+0.08, +0.23)	+0.00 (-0.05, +0.06)	+0.01 (-0.01, +0.02)
Limb amplitude change	+0.39 (+0.29, +0.49)	+0.29 (+0.19, +0.39)	+0.35 (+0.28, +0.43)	+0.03 (+0.01, +0.05)
Precordial amplitude change	+0.58 (+0.49, +0.67)	+0.43 (+0.34, +0.52)	+0.39 (+0.32, +0.45)	+0.16 (+0.14, +0.18)
Interval (log minutes)	+0.05 (-0.02, +0.11)	-0.05 (-0.12, +0.02)	-0.03 (-0.08, +0.02)	+0.00 (-0.01, +0.02)
Chronological age	-0.43 (-0.50, -0.36)	-0.38 (-0.46, -0.30)	-0.47 (-0.53, -0.42)	-0.08 (-0.10, -0.07)
Male (binary)	-0.02 (-0.15, +0.12)	-0.23 (-0.38, -0.08)	-0.24 (-0.34, -0.13)	-0.15 (-0.18, -0.12)
Device change (binary)	-0.34 (-0.49, -0.19)	-0.14 (-0.30, +0.02)	-0.19 (-0.31, -0.08)	+0.02 (-0.02, +0.05)
Filter change (binary)	+0.17 (-0.01, +0.34)	+0.29 (+0.09, +0.49)	+0.32 (+0.18, +0.46)	+0.06 (+0.02, +0.11)
Rhythm-class change (binary)	+1.11 (+0.89, +1.34)	+1.31 (+1.06, +1.56)	+0.99 (+0.81, +1.16)	+0.03 (-0.01, +0.08)

Predictor	Lima: abs. change in age gap, y	Bracke: abs. change in age gap, y	In-house: abs. change in age gap, y	Sex: abs. change in logit
Paced (binary)	-1.13 (-1.56, -0.70)	-0.27 (-0.76, +0.22)	-0.32 (-0.70, +0.05)	-0.21 (-0.30, -0.11)
QRS-duration difference	+0.01 (-0.10, +0.13)	+0.02 (-0.01, +0.06)	-0.02 (-0.03, +0.00)	-0.00 (-0.01, +0.00)
<i>n</i> ; <i>R</i> ²	22 512; 0.077	22 512; 0.053	22 512; 0.071	22 512; 0.041
Mean outcome	5.66	6.08	4.23	1.41

Table S6d. MIMIC ED, troponin-negative, 1–6 h.

Predictor	Lima: abs. change in age gap, y	Bracke: abs. change in age gap, y	In-house: abs. change in age gap, y	Sex: abs. change in logit
Heart-rate difference	+0.50 (+0.30, +0.70)	+0.20 (+0.03, +0.38)	+0.17 (+0.04, +0.30)	+0.06 (+0.01, +0.10)
RR variability (higher of pair)	-0.18 (-0.34, -0.02)	+0.01 (-0.14, +0.17)	-0.11 (-0.22, +0.00)	-0.09 (-0.12, -0.05)
High-frequency noise (higher of pair, log)	-0.01 (-0.15, +0.13)	+0.03 (-0.11, +0.17)	+0.02 (-0.09, +0.12)	-0.00 (-0.04, +0.04)
Baseline wander (higher of pair, log)	-0.02 (-0.17, +0.13)	+0.18 (+0.03, +0.33)	+0.07 (-0.05, +0.18)	+0.05 (+0.01, +0.09)
Limb amplitude change	+0.29 (+0.09, +0.48)	+0.23 (+0.05, +0.42)	+0.26 (+0.11, +0.42)	+0.01 (-0.04, +0.05)
Precordial amplitude change	+0.35 (+0.15, +0.55)	+0.26 (+0.09, +0.44)	+0.19 (+0.06, +0.32)	+0.16 (+0.11, +0.21)
Interval (log minutes)	-0.07 (-0.22, +0.08)	+0.04 (-0.12, +0.19)	-0.03 (-0.15, +0.09)	+0.04 (+0.00, +0.08)
Chronological age	-0.01 (-0.15, +0.12)	-0.06 (-0.20, +0.08)	-0.38 (-0.49, -0.26)	-0.04 (-0.08, -0.00)
Male (binary)	+0.13 (-0.16, +0.41)	-0.20 (-0.49, +0.09)	-0.38 (-0.60, -0.16)	-0.14 (-0.21, -0.06)
Device change (binary)	-0.04 (-0.39, +0.31)	+0.20 (-0.18, +0.58)	-0.12 (-0.40, +0.15)	+0.00 (-0.10, +0.10)
Filter change (binary)	-0.20 (-0.76, +0.35)	-0.24 (-0.83, +0.36)	-0.04 (-0.47, +0.39)	-0.10 (-0.25, +0.05)
Rhythm-class change (binary)	+0.95 (-0.38, +2.28)	+1.42 (+0.16, +2.68)	+0.69 (-0.23, +1.60)	-0.20 (-0.45, +0.05)
Paced (binary)	+0.00 (+0.00, +0.00)	+0.00 (+0.00, +0.00)	+0.00 (+0.00, +0.00)	+0.00 (+0.00, +0.00)
QRS-duration difference	+0.29 (+0.03, +0.55)	+0.12 (-0.10, +0.34)	+0.25 (+0.08, +0.43)	+0.00 (-0.04, +0.05)
<i>n</i> ; <i>R</i> ²	3 780; 0.040	3 780; 0.023	3 780; 0.041	3 780; 0.038
Mean outcome	5.07	5.23	3.89	1.43

Table S7a. Pre-specified sensitivity analyses at 1–6 hours.

Cohort	Analysis	<i>n</i>	Lima <i>S_w</i>	Bracke <i>S_w</i>	In-house <i>S_w</i>	Sex class change, %
I0001	All pairs (primary)	17 409	6.52	6.79	4.89	14.3
I0001	Regular rhythm (RR CV ≤ 0.15)	12 995	6.43	6.56	4.82	12.8
I0001	Adults only (sex)	20 412	—	—	—	13.9
I0006	All pairs (primary)	21 507	6.07	6.53	4.59	15.3
I0006	Regular rhythm (RR CV ≤ 0.15)	15 479	6.08	6.50	4.60	14.7
I0006	Adults only (sex)	22 565	—	—	—	15.3
MIMIC	All pairs (primary)	22 532	5.39	5.84	4.11	13.0
MIMIC	Regular rhythm (RR CV ≤ 0.15)	15 941	5.32	5.73	4.12	11.8
MIMIC	Same device and filter	9 299	5.67	6.00	4.26	12.7

Cohort	Analysis	n	Lima S_w	Bracke S_w	In-house S_w	Sex class change, %
MIMIC	Adults only (sex)	24 107	—	—	—	13.0
MIMIC ED	All pairs (primary)	3 780	4.73	4.85	3.67	8.7
MIMIC ED	Regular rhythm (RR CV ≤ 0.15)	3 311	4.70	4.80	3.69	8.2
MIMIC ED	Same device and filter	2 797	4.81	4.84	3.74	8.8
MIMIC ED	Adults only (sex)	3 906	—	—	—	8.7

Table S7b. Age-gap S_w (Lima model, years) after excluding pairs with artifact or electrode-reversal flags (post hoc).

Cohort	Interval	All	Excl. noisy (top 10%)	Excl. MIMIC machine flags	Excl. signal detector	Excl. 12SL + detector	Detector flagged, %	12SL + detector flagged, %	Flagged pairs only
I0001	<5 min	3.73	3.56	—	3.57	3.51	1.9	6.1	8.47
I0001	1–6 h	6.53	6.41	—	6.39	6.36	3.8	6.5	9.43
I0001	>1 y	7.53	7.46	—	7.33	7.29	4.7	6.2	10.76
I0006	<5 min	4.05	3.98	—	3.91	3.86	2.2	5.0	8.24
I0006	1–6 h	6.07	5.97	—	5.92	5.91	4.0	5.1	8.90
I0006	>1 y	7.36	7.20	—	7.09	7.10	5.7	6.2	10.85
MIMIC	<5 min	4.24	4.28	4.18	4.18	—	2.0	—	6.53
MIMIC	1–6 h	5.39	5.31	5.36	5.32	—	2.4	—	7.77
MIMIC	>1 y	6.08	6.03	6.07	6.03	—	2.0	—	8.37
MIMIC ED	1–6 h	4.73	4.64	4.69	4.73	—	1.6	—	4.82

Table S8. Validation of the pairwise electrode-reversal detector. Reference: pairs in which exactly one ECG carried a machine reversal statement.

Reference	Pairs with reversal in one ECG	AUC	Limb threshold	Flagged, %	Sensitivity, %	Positive predictive value, %
MIMIC machine statement	1 174	0.90	0.5	0.44	32.0	50.9
MIMIC machine statement	1 174	0.90	0.6	0.63	41.1	45.7
MIMIC machine statement	1 174	0.90	0.7	0.96	48.8	35.5
MIMIC machine statement	1 174	0.90	0.8	1.61	54.6	23.7
HEEDB 12SL v24	5 690	0.69	0.5	0.71	15.1	33.2
HEEDB 12SL v24	5 690	0.69	0.6	1.23	19.0	24.1
HEEDB 12SL v24	5 690	0.69	0.7	2.18	23.2	16.6
HEEDB 12SL v24	5 690	0.69	0.8	3.86	28.0	11.3

Table S9a. Signal change between the two ECGs by interval (post hoc).

Cohort	<5 min	5–60 min	1–6 h	6–24 h	1–7 d	7–30 d	30 d–1 y	>1 y
I0001	9 / 8 / 2.0	19 / 17 / 5.6	23 / 20 / 7.6	26 / 23 / 9.0	30 / 25 / 10.5	30 / 25 / 10.4	28 / 22 / 9.1	33 / 26 / 10.0
I0006	11 / 10 / 2.3	24 / 22 / 9.2	25 / 21 / 7.2	28 / 23 / 8.8	31 / 26 / 11.3	31 / 26 / 10.4	29 / 24 / 8.6	28 / 24 / 7.8

Cohort	<5 min	5–60 min	1–6 h	6–24 h	1–7 d	7–30 d	30 d–1 y	>1 y
MIMIC	10 / 10 / 2.7	17 / 12 / 5.3	21 / 15 / 7.2	24 / 16 / 8.4	28 / 19 / 10.8	28 / 18 / 10.5	26 / 17 / 9.3	26 / 17 / 9.8

Cells: median precordial / limb amplitude change (mean $|\log \text{RMS ratio}| \times 100$) / median absolute heart-rate difference (b.p.m.).

Table S9b. 1–6-hour pairs whose signal change was below the median of <5-minute pairs on all three measures (post hoc).

Cohort	n	Lima S_w	Bracke S_w	In-house S_w	Lima S_w <5 min	Bracke S_w <5 min
I0001	185	3.52	4.63	3.01	3.74	4.16
I0006	266	3.72	4.67	3.29	4.06	4.17
MIMIC	542	3.53	3.98	2.80	4.25	4.54

Table S10. Averaging two ECGs in patients with three consecutive ECGs within 6 hours.

Cohort	Output	Patients	RMS difference, single ECG	RMS difference, mean of two	Ratio (expected 0.866)
I0001	Lima age	18 164	9.03	7.47	0.827
I0001	Bracke age	18 163	9.45	7.91	0.837
I0001	In-house age	18 164	6.78	5.66	0.834
I0001	In-house sex logit	18 164	1.76	1.47	0.834
I0006	Lima age	6 465	9.65	8.14	0.843
I0006	Bracke age	6 454	9.94	8.43	0.848
I0006	In-house age	6 465	7.19	6.18	0.859
I0006	In-house sex logit	6 465	2.05	1.72	0.839
MIMIC	Lima age	6 815	8.35	7.13	0.854
MIMIC	Bracke age	6 815	9.10	7.77	0.854
MIMIC	In-house age	6 815	6.34	5.45	0.859
MIMIC	In-house sex logit	6 815	1.86	1.59	0.854

Table S11. Published abnormality classifier: all classes. AI call: probability >0.5. “Lost”: among patients positive on the first ECG, the proportion not positive on the second.

Cohort	Interval	Class	AI positive (first ECG), %	AI call lost, % (95% CI)	Machine positive, %	Machine statement lost, %	κ AI	κ machine	Logit ICC
I0001	<5 min	First-degree AV block	2.2	15.6 (12.6–18.6)	5.5	24.2	0.82	0.72	0.95
I0001	<5 min	RBBB	4.3	6.0 (4.6–7.4)	5.4	8.0	0.92	0.91	0.95
I0001	<5 min	LBBB	2.1	6.7 (4.5–8.8)	2.2	13.0	0.92	0.85	0.96
I0001	<5 min	Sinus bradycardia	1.6	25.6 (21.2–30.1)	11.0	19.5	0.72	0.76	0.90
I0001	<5 min	Atrial fibrillation	7.6	12.2 (10.7–13.6)	8.3	24.2	0.86	0.75	0.93
I0001	<5 min	Sinus tachycardia	8.2	13.4 (12.0–15.0)	10.8	19.3	0.85	0.79	0.92
I0001	1–6 h	First-degree AV block	2.2	39.3 (35.3–43.6)	5.7	39.9	0.56	0.56	0.82

Cohort	Interval	Class	AI positive (first ECG), %	AI call lost, % (95% CI)	Machine positive, %	Machine statement lost, %	κ AI	κ machine	Logit ICC
I0001	1–6 h	RBBB	4.5	18.4 (16.2–20.7)	5.7	20.8	0.82	0.80	0.86
I0001	1–6 h	LBBB	2.6	17.9 (14.9–21.2)	2.4	25.7	0.81	0.74	0.90
I0001	1–6 h	Sinus bradycardia	1.3	48.3 (43.0–53.9)	8.8	40.3	0.43	0.50	0.68
I0001	1–6 h	Atrial fibrillation	9.7	37.4 (35.3–39.3)	10.5	42.1	0.64	0.57	0.75
I0001	1–6 h	Sinus tachycardia	11.0	46.1 (44.2–48.0)	14.3	47.5	0.55	0.52	0.62
I0001	>1 y	First-degree AV block	1.5	51.6 (46.4–56.8)	3.6	51.0	0.39	0.39	0.63
I0001	>1 y	RBBB	2.0	17.7 (14.5–21.2)	2.3	18.6	0.62	0.60	0.64
I0001	>1 y	LBBB	1.0	22.3 (17.2–27.7)	1.0	26.6	0.59	0.56	0.71
I0001	>1 y	Sinus bradycardia	1.5	76.5 (72.1–80.9)	14.4	59.7	0.20	0.30	0.41
I0001	>1 y	Atrial fibrillation	2.1	29.4 (25.2–33.6)	2.7	42.3	0.46	0.41	0.57
I0001	>1 y	Sinus tachycardia	3.7	76.7 (73.9–79.7)	5.0	78.0	0.16	0.15	0.26
I0006	<5 min	First-degree AV block	2.6	14.1 (10.9–17.5)	6.5	23.2	0.82	0.74	0.94
I0006	<5 min	RBBB	4.8	5.7 (4.1–7.4)	6.0	9.0	0.93	0.89	0.95
I0006	<5 min	LBBB	2.0	8.5 (5.7–11.6)	2.0	14.5	0.88	0.84	0.96
I0006	<5 min	Sinus bradycardia	2.0	34.3 (29.1–39.8)	14.2	21.0	0.67	0.75	0.87
I0006	<5 min	Atrial fibrillation	7.0	14.7 (12.6–16.7)	6.4	22.9	0.85	0.76	0.92
I0006	<5 min	Sinus tachycardia	6.3	19.9 (17.4–22.4)	8.2	25.0	0.77	0.72	0.88
I0006	1–6 h	First-degree AV block	2.8	31.5 (27.8–35.2)	7.1	31.6	0.57	0.57	0.82
I0006	1–6 h	RBBB	6.4	16.9 (15.0–18.9)	7.5	20.4	0.84	0.79	0.86
I0006	1–6 h	LBBB	3.4	22.0 (18.9–25.0)	3.1	29.9	0.77	0.69	0.90
I0006	1–6 h	Sinus bradycardia	1.4	57.4 (51.6–62.8)	11.3	36.0	0.34	0.53	0.63
I0006	1–6 h	Atrial fibrillation	13.0	50.5 (48.7–52.3)	10.8	52.4	0.52	0.50	0.72
I0006	1–6 h	Sinus tachycardia	10.0	45.2 (43.2–47.3)	13.3	44.4	0.54	0.53	0.65
I0006	>1 y	First-degree AV block	3.2	51.3 (47.7–54.9)	6.7	44.4	0.46	0.50	0.75
I0006	>1 y	RBBB	4.7	20.0 (17.8–22.4)	5.4	20.4	0.74	0.73	0.76
I0006	>1 y	LBBB	2.1	29.8 (25.8–33.9)	1.9	35.5	0.65	0.58	0.81
I0006	>1 y	Sinus bradycardia	2.8	70.4 (66.8–73.8)	20.0	49.6	0.28	0.39	0.49
I0006	>1 y	Atrial	4.1	32.9 (30.0–35.8)	3.6	30.3	0.55	0.56	0.67

Cohort	Interval	Class	AI positive (first ECG), %	AI call lost, % (95% CI)	Machine positive, %	Machine statement lost, %	κ AI	κ machine	Logit ICC
I0006	>1 y	fibrillation	3.2	80.1 (77.2–82.9)	3.8	80.4	0.17	0.16	0.29
MIMIC	<5 min	Sinus tachycardia	3.1	22.1 (16.2–27.9)	6.7	39.5	0.72	0.54	0.90
MIMIC	<5 min	First-degree AV block	7.0	10.9 (8.3–13.8)	5.8	20.1	0.86	0.79	0.92
MIMIC	<5 min	RBBB	3.6	15.0 (10.7–19.7)	3.3	22.9	0.81	0.75	0.93
MIMIC	<5 min	LBBB	1.3	21.7 (13.3–31.3)	9.7	24.3	0.63	0.68	0.81
MIMIC	<5 min	Sinus bradycardia	15.6	17.8 (15.5–20.2)	18.4	35.7	0.78	0.59	0.87
MIMIC	<5 min	Atrial fibrillation	6.2	23.5 (19.6–27.9)	9.0	32.7	0.73	0.62	0.83
MIMIC	1–6 h	Sinus tachycardia	3.0	33.0 (29.6–36.5)	6.1	41.6	0.57	0.45	0.83
MIMIC	1–6 h	First-degree AV block	5.8	13.1 (11.5–14.9)	4.7	25.6	0.87	0.75	0.88
MIMIC	1–6 h	RBBB	3.6	18.3 (15.6–20.9)	3.2	27.2	0.83	0.76	0.92
MIMIC	1–6 h	LBBB	1.8	51.4 (46.8–55.9)	11.0	37.5	0.40	0.47	0.64
MIMIC	1–6 h	Sinus bradycardia	14.2	47.0 (45.4–48.7)	14.1	50.4	0.57	0.51	0.73
MIMIC	1–6 h	Atrial fibrillation	9.5	50.2 (48.2–52.1)	12.0	50.8	0.53	0.50	0.62
MIMIC	>1 y	Sinus tachycardia	3.1	49.5 (46.1–53.2)	7.0	57.1	0.49	0.41	0.76
MIMIC	>1 y	First-degree AV block	3.9	13.3 (11.1–15.5)	3.1	25.9	0.78	0.66	0.77
MIMIC	>1 y	RBBB	1.7	18.4 (14.9–22.2)	1.6	26.0	0.73	0.63	0.84
MIMIC	>1 y	LBBB	2.2	77.0 (73.7–80.4)	16.8	62.1	0.22	0.30	0.43
MIMIC	>1 y	Sinus bradycardia	3.8	24.6 (22.0–27.3)	4.1	28.4	0.56	0.54	0.66
MIMIC	>1 y	Atrial fibrillation	4.7	76.2 (73.8–78.7)	5.6	74.4	0.17	0.18	0.24
MIMIC ED	1–6 h	Sinus tachycardia	2.4	25.0 (16.3–33.7)	3.3	33.9	0.67	0.57	0.85
MIMIC ED	1–6 h	First-degree AV block	3.5	8.1 (3.7–12.6)	2.8	18.2	0.93	0.86	0.87
MIMIC ED	1–6 h	RBBB	1.4	5.6 (0.0–13.0)	1.6	15.9	0.89	0.83	0.93
MIMIC ED	1–6 h	LBBB	1.8	54.3 (42.9–65.7)	12.9	33.0	0.31	0.47	0.64
MIMIC ED	1–6 h	Sinus bradycardia	3.7	8.2 (4.1–13.0)	3.3	6.3	0.92	0.92	0.83
MIMIC ED	1–6 h	Atrial fibrillation	5.3	70.5 (64.3–76.8)	6.5	64.3	0.36	0.42	0.53
MIMIC ED	1–6 h	Sinus tachycardia							

Table S12. GRRAS checklist.

Item	GRRAS recommendation	Location
1	Identify in title/abstract that reliability was investigated	Title, Abstract
2	Name and describe the measured outcome	Methods: AI models; Table 2
3	Specify the subject population	Methods: Data sources; Table 1
4	Specify the rater population	Not applicable: automated models (Methods: AI models)
5	Describe what was known about reliability before	Introduction
6	Explain how subjects were selected	Methods: pairs and strata; Supplementary Methods S1
7	Describe sample size considerations	Methods (all eligible pairs, capped at 25 000 per stratum)
8	Describe the measurement and rating process	Methods; Supplementary Methods S1–S2
9	State whether measurements were independent	Methods: models frozen; outputs computed per ECG
10	Describe the statistical analysis	Methods; Supplementary Methods S5
11	State the actual number of subjects and replicates	Results; Table S1
12	Describe the sample characteristics	Table 1
13	Report estimates with measures of statistical uncertainty	Tables 2–4; Tables S2–S11
14	Discuss the practical relevance of results	Discussion: Implications
15	Provide detailed results if possible	Supplementary material

Supplementary Figures

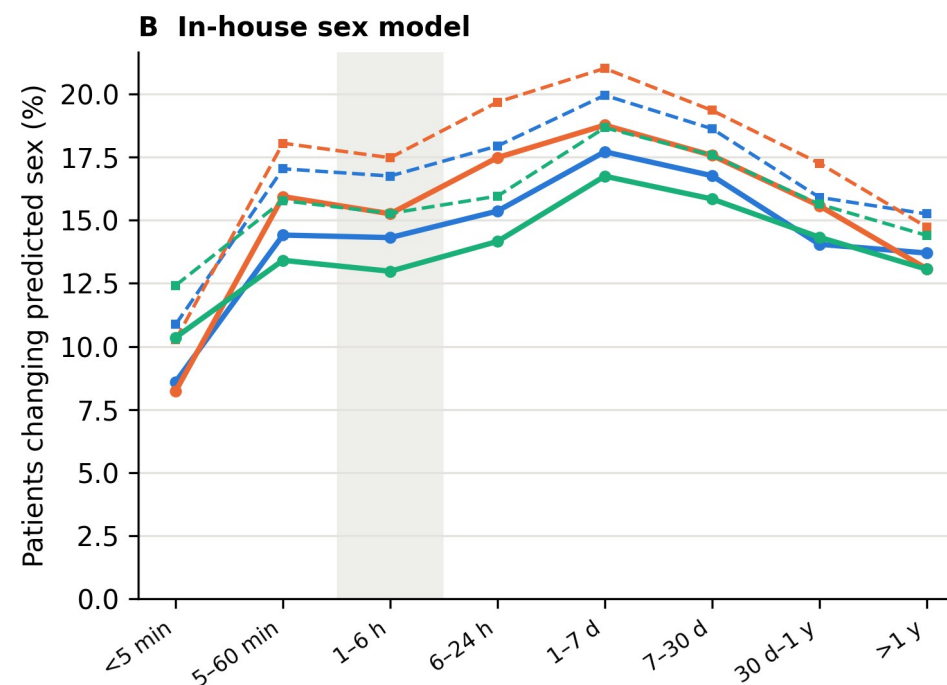
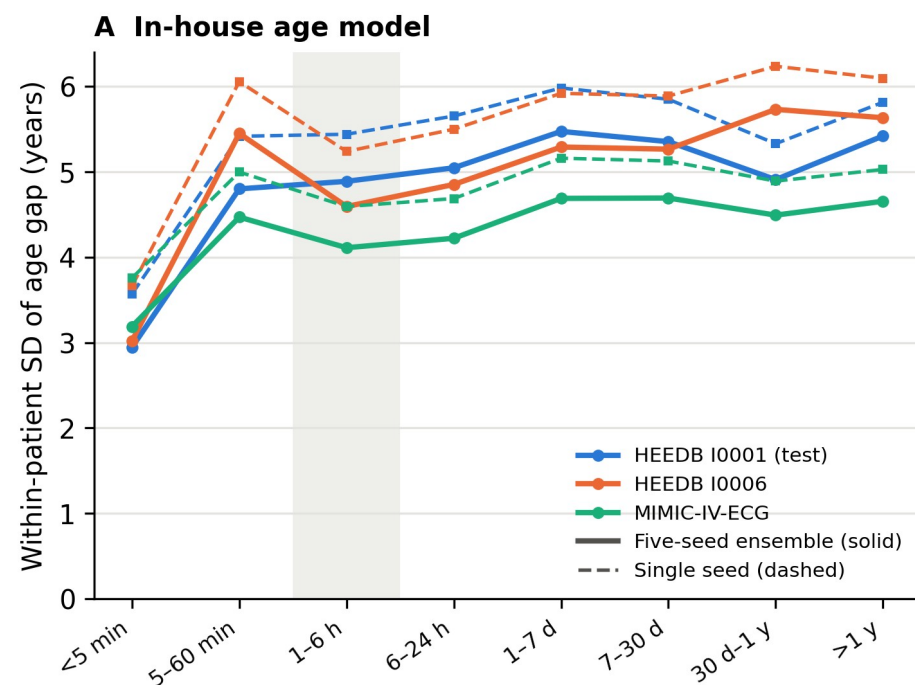


Figure S1. In-house models by interval. (A) Within-patient SD of the age gap for the five-seed ensemble (solid) and a single seed (dashed). (B) Percentage of patients whose predicted sex changed, for the ensemble (solid) and a single seed (dashed). The grey band marks the primary 1–6-hour stratum.

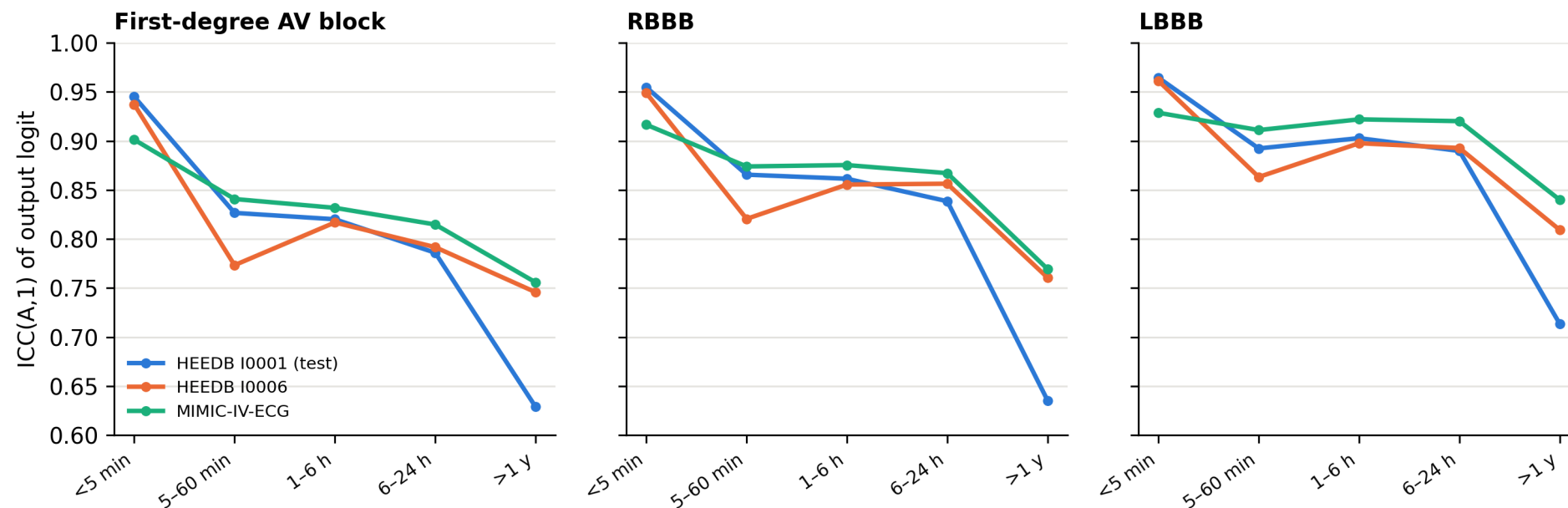


Figure S2. Published abnormality classifier: ICC(A,1) of the output logit for conduction abnormalities by interval.

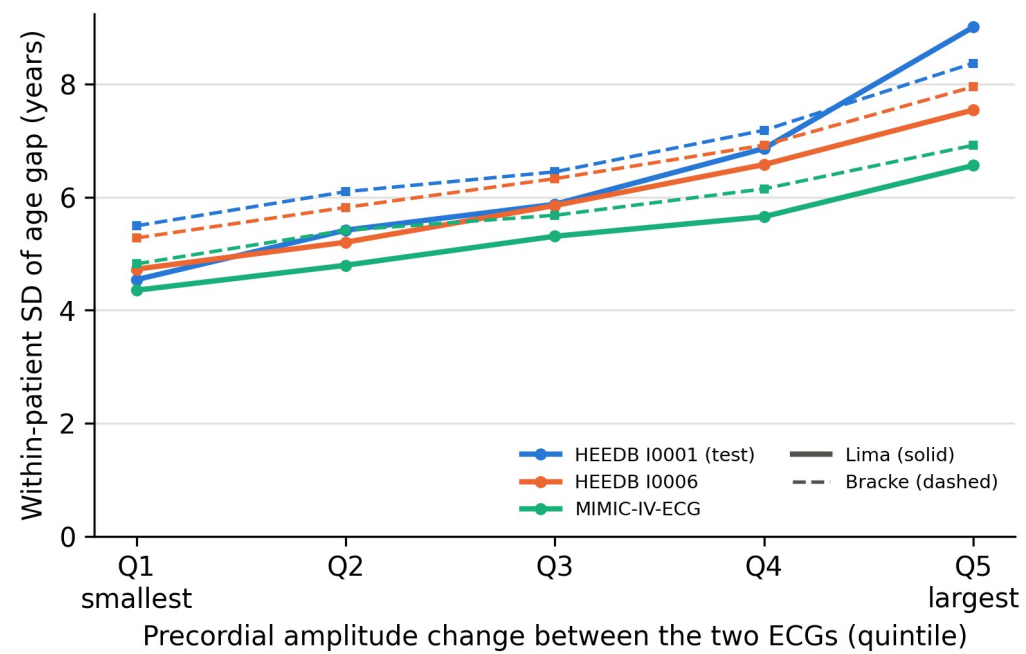


Figure S3. Age-gap variability in 1–6-hour pairs by quintile of precordial amplitude change, for both published ECG-age models (post hoc).